Foundation Mathematics 1017SCG
Week 4 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\begin{aligned} 3(x+6) &= 3 \times x + 3 \times 6 \\ &= 3x + 18 \end{aligned}$$

(b)

$$\begin{aligned} 2(2y+4) &= 2 \times 2y + 2 \times 4 \\ &= 4y + 8 \end{aligned}$$

(c)

$$\begin{aligned} 5x(2x-3) &= 5x \times 2x + 5x \times -3 \\ &= 10x^2 - 15x \end{aligned}$$

(d)

$$\begin{aligned} 4y(5y-1) &= 4y \times 5y + 4y \times -1 \\ &= 20y^2 - 4y \end{aligned}$$

(e)

$$\begin{aligned} 7x(2x+3) - 5x &= 7x \times 2x + 7x \times 3 - 5x \\ &= 14x^2 + 21x - 5x \\ &= 14x^2 + 16x \end{aligned}$$

(f)

$$\begin{aligned} -3y(5y-2) + 4 &= -3y \times 5y - 3y \times -2 + 4 \\ &= -15y^2 + 6y + 4 \end{aligned}$$

(g) To assist in expanding this question, you may find it helpful to rewrite the question with a '-1' in front of the second bracket.

$$\begin{aligned} -2(n-4) - (3n+1) &= -2(n-4) - 1(3n+1) \\ &= -2 \times n - 2 \times -4 - 1 \times 3n - 1 \times 1 \\ &= -2n + 8 - 3n - 1 \\ &= -5n + 7 \end{aligned}$$

(h)

$$\begin{aligned} x(2x-3) + 4(x^2+2) &= x \times 2x + x \times -3 + 4 \times x^2 + 4 \times 2 \\ &= 2x^2 - 3x + 4x^2 + 8 \\ &= 6x^2 - 3x + 8 \end{aligned}$$

(i)

$$\begin{aligned}2a(5a-1)+3(4a+2) &= 2a\times 5a+2a\times -1+3\times 4a+3\times 2\\ &= 10a^2-2a+12a+6\\ &= 10a^2+10a+6\end{aligned}$$

(j)

$$\begin{aligned}2g(1-2g)-5(g-2) &= 2g\times 1+2g\times -2g-5\times g-5\times -2\\ &= 2g-4g^2-5g+10\\ &= -4g^2-3g+10\end{aligned}$$

(k)

$$\begin{aligned}4(x-3)-2(x-2) &= 4\times x+4\times -3-2\times x-2\times -2\\ &= 4x-12-2x+4\\ &= 2x-8\end{aligned}$$

(l)

$$\begin{aligned}12(n-3)+n(-2-2n) &= 12\times n+12\times -3+n\times -2+n\times -2n\\ &= 12n-36-2n-2n^2\\ &= -2n^2+10n-36\end{aligned}$$

2. (a)

$$\begin{aligned}(a+4)^2 &= (a+4)(a+4)\\ &= a\times a+a\times 4+4\times a+4\times 4\\ &= a^2+4a+4a+16\\ &= a^2+8a+16\end{aligned}$$

(b)

$$\begin{aligned}(2x-3)^2 &= (2x-3)(2x-3)\\ &= 2x\times 2x+2x\times -3-3\times 2x-3\times -3\\ &= 4x^2-6x-6x+9\\ &= 4x^2-12x+9\end{aligned}$$

(c)

$$\begin{aligned}(y-7)^2 &= (y-7)(y-7)\\ &= y\times y+y\times -7-7\times y-7\times -7\\ &= y^2-7y-7y+49\\ &= y^2-14y+49\end{aligned}$$

(d)

$$\begin{aligned}(n+6)(n-6) &= n\times n+n\times -6+6\times n+6\times -6\\ &= n^2-6n+6n-36\\ &= n^2-36\end{aligned}$$

(e)

$$\begin{aligned}(y+2)(y-2) &= y \times y + y \times -2 + 2 \times y + 2 \times -2 \\ &= y^2 - 2y + 2y - 4 \\ &= y^2 - 4\end{aligned}$$

(f)

$$\begin{aligned}(3x+4)(3x-4) &= 3x \times 3x + 3x \times -4 + 4 \times 3x + 4 \times -4 \\ &= 9x^2 - 12x + 12x - 16 \\ &= 9x^2 - 16\end{aligned}$$

3. (a)

$$\begin{aligned}(x+2)(x+3) &= x^2 + 3x + 2x + 6 \\ &= x^2 + 5x + 6\end{aligned}$$

(b)

$$\begin{aligned}(x+3)(x+8) &= x^2 + 8x + 3x + 24 \\ &= x^2 + 11x + 24\end{aligned}$$

(c)

$$\begin{aligned}(x-4)(x-5) &= x^2 - 5x - 4x + 20 \\ &= x^2 - 9x + 20\end{aligned}$$

(d)

$$\begin{aligned}(2y+7)(4y-1) &= 8y^2 - 2y + 28y - 7 \\ &= 8y^2 + 26y - 7\end{aligned}$$

(e)

$$\begin{aligned}(3b+8)(10b-3) &= 30b^2 - 9b + 80b - 24 \\ &= 30b^2 + 71b - 24\end{aligned}$$

(f)

$$\begin{aligned}(7h-2)(3h-5) &= 21h^2 - 35h - 6h + 10 \\ &= 21h^2 - 41h + 10\end{aligned}$$

(g)

$$\begin{aligned}(a-2)(a+5) &= a^2 + 5a - 2a - 10 \\ &= a^2 + 3a - 10\end{aligned}$$

(h)

$$\begin{aligned}(2x-1)(3x+4) &= 6x^2 + 8x - 3x - 4 \\ &= 6x^2 + 5x - 4\end{aligned}$$

4. (a)

$$6x + 2 = 2(3x + 1)$$

(b)

$$5x - 10 = 5(x - 2)$$

(c)

$$x^2 - 8x = x(x - 8)$$

(d)

$$2x^3 + 7x^2 = x^2(2x + 7)$$

(e)

$$6y^2 + 9y = 3y(2y + 3)$$

(f)

$$10y^5 - 25y^3 = 5y^3(2y^2 - 5)$$

(g)

$$4a^3 + 12a^2 = 4a^2(a + 3)$$

(h)

$$2x^3y^2 - 4x^3y^5 = 2x^3y^2(1 - 2y^3)$$

(i)

$$14x^2y^2 + 21x^4y = 7x^2y(2y + 3x^2)$$

(j)

$$3a^2b - 15ab^2 = 3ab(a - 5b)$$

5. (a)

$$\begin{aligned} x^2 - 25 &= x^2 - 5^2 \\ &= (x + 5)(x - 5) \end{aligned}$$

(b)

$$\begin{aligned} y^2 - 9 &= y^2 - 3^2 \\ &= (y + 3)(y - 3) \end{aligned}$$

(c)

$$\begin{aligned} x^2 - 16 &= x^2 - 4^2 \\ &= (x + 4)(x - 4) \end{aligned}$$

(d)

$$\begin{aligned} y^2 - 1 &= y^2 - 1^2 \\ &= (y + 1)(y - 1) \end{aligned}$$

(e)

$$\begin{aligned} 4x^2 - 81 &= (2x)^2 - 9^2 \\ &= (2x + 9)(2x - 9) \end{aligned}$$

(f)

$$\begin{aligned} 100y^2 - 49 &= (10y)^2 - 7^2 \\ &= (10y+7)(10y-7) \end{aligned}$$

(g)

$$\begin{aligned} 5x^2 - 20 &= 5(x^2 - 4) \\ &= 5(x^2 - 2^2) \\ &= 5(x+2)(x-2) \end{aligned}$$

(h)

$$\begin{aligned} 3y^2 - 75 &= 3(y^2 - 25) \\ &= 3(y^2 - 5^2) \\ &= 3(y+5)(y-5) \end{aligned}$$

6. In this question, you will need to factorise quadratics using the 'cross method'. It is important to remember that you will not always guess the correct factors straight away. It may take a few tries before you find the right factors.

   (a)
   $$x^2 + 5x + 6 = (x+2)(x+3)$$

   (b)
   $$y^2 + 11y + 10 = (y+1)(y+10)$$

   (c)
   $$x^2 + 6x + 8 = (x+2)(x+4)$$

   (d)
   $$y^2 - 5y + 4 = (y-1)(y-4)$$

   (e)
   $$x^2 - 13x + 12 = (x-1)(x-12)$$

   (f)
   $$a^2 - 11a - 12 = (a-12)(a+1)$$

   (g)
   $$y^2 - 9y + 20 = (y-4)(y-5)$$

   (h) It is important to note that in this question, there is a common factor. By first factorising using a common factor, it will make the cross method easier.

   $$\begin{aligned} 2n^2 + 14n + 20 &= 2(n^2 + 7n + 10) \\ &= 2(n+5)(n+2) \end{aligned}$$

   (i)

   $$\begin{aligned} 3k^2 + 33k + 30 &= 3(k^2 + 11k + 10) \\ &= 3(k+1)(k+10) \end{aligned}$$

(j) As all terms are negative, first factoring using a common factor of -1 will make the cross method easier.

$$\begin{aligned} -x^2 - 8x - 12 &= -(x^2 + 8x + 12) \\ &= -(x + 2)(x + 6) \end{aligned}$$

7. (a)

$$2x^2 - 11x + 5 = (2x - 1)(x - 5)$$

(b)

$$2x^2 + 5x + 3 = (2x + 3)(x + 1)$$

(c)

$$3x^2 + 14x + 8 = (3x + 2)(x + 4)$$

(d)

$$3x^2 - 8x + 5 = (3x - 5)(x - 1)$$

(e)

$$4x^2 + 9x + 2 = (4x + 1)(x + 2)$$

(f)

$$4x^2 - 8x + 3 = (2x - 1)(2x - 3)$$

8. (a)

$$\begin{aligned} \frac{x}{25} \times \frac{10}{y} &= \frac{x}{5} \times \frac{2}{y} \\ &= \frac{2x}{5y} \end{aligned}$$

(b)

$$\begin{aligned} \frac{20}{21a} \times \frac{7c}{10} &= \frac{2}{3a} \times \frac{c}{1} \\ &= \frac{2c}{3a} \end{aligned}$$

(c)

$$\begin{aligned} \frac{3nm}{5} \div \frac{9n}{10} &= \frac{3nm}{5} \times \frac{10}{9n} \\ &= \frac{m}{1} \times \frac{2}{3} \\ &= \frac{2m}{3} \end{aligned}$$

(d)

$$\begin{aligned} \frac{x - 7}{1 - x^2} \div \frac{2x - 14}{x + 1} &= \frac{x - 7}{1 - x^2} \times \frac{x + 1}{2x - 14} \\ &= \frac{x - 7}{(1 + x)(1 - x)} \times \frac{x + 1}{2(x - 7)} \\ &= \frac{1}{1 - x} \times \frac{1}{2} \\ &= \frac{1}{2(1 - x)} \end{aligned}$$

(e)

$$\begin{aligned}\frac{3y}{(2y^2-y-10)}\times\frac{2y-5}{9y^2} &= \frac{3y}{(2y-5)(y+2)}\times\frac{2y-5}{9y^2}\\ &= \frac{1}{y+2}\times\frac{1}{3y}\\ &= \frac{1}{3y(y+2)}\end{aligned}$$

(f)

$$\begin{aligned}\frac{4a}{a^2+a-6}\div\frac{2a}{a^2-9} &= \frac{4a}{a^2+a-6}\times\frac{a^2-9}{2a}\\ &= \frac{2}{a-2}\times\frac{a-3}{1}\\ &= \frac{2(a-3)}{a-2}\end{aligned}$$

(g)

$$\begin{aligned}\frac{2x^2-5x-3}{x^2-4}\div\frac{x^2-3x}{2x+4} &= \frac{2x^2-5x-3}{x^2-4}\times\frac{2x+4}{x^2-3x}\\ &= \frac{(2x+1)(x-3)}{(x+2)(x-2)}\times\frac{2(x+2)}{x(x-3)}\\ &= \frac{(2x+1)}{(x-2)}\times\frac{2}{x}\\ &= \frac{2(2x+1)}{x(x-2)}\end{aligned}$$

9. (a)

$$\begin{aligned}(2x+3)(x-4) &= 2x^2-8x+3x-12\\ &= 2x^2-5x-12\end{aligned}$$

(b)

$$\begin{aligned}(2x+3)(x-4)(x+1) &= (2x^2-5x-12)(x+1)\\ &= 2x^3+2x^2-5x^2-5x-12x-12\\ &= 2x^3-3x^2-17x-12\end{aligned}$$